Context is the New Weight: Distilling In-Context Adaptation into Weight Updates and Comparing the Mechanism at the Attention Layer

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Abstract

We compare two ways of making a language model behave according to a context C: prepending C at inference time (**in-context**) and fine-tuning the model on synthetic (Q, A) pairs generated under C (**context-distilled**), both evaluated against the unmodified **base** model. We compare *activations*, layer by layer, on Llama-3.1-8B-Instruct across seven context types — structure constraints, style constraints, few-shot demonstrations, factual injection, conversational history, and a compressed-history summary. We find that context-distilled produces a residual-stream perturbation aligned with the in-context perturbation in direction (cosine 0.42–0.80 at peak sites in the last third of the network) but with consistently larger magnitude. The activation pattern is invariant across training and held-out validation queries, ruling out memorisation. Crucially, evaluating both variants on the unrelated MMLU-Pro benchmark reveals that in-context preserves general capability (Δ acc $\in [-0.07, +0.02]$ across all seven contexts), while context-distilled destroys it for most contexts (Δ acc as low as -0.24). We test prefix tuning as a candidate fix — preserving the "attention reads extra K/V" mechanism while internalising the context with only $\approx 1\,\mathrm{M}$ trainable parameters — and find that it also destroys MMLU-Pro (Δ acc as low as -0.29), and in some cases more severely than full distillation. The shared failure mode is that any bake-in is *always active* on every forward pass, while in-context is content-conditional via attention to C's actual tokens. Compressing context into model parameters — whether the whole model or a tiny adapter — is therefore not a clean substitute for keeping the context in the input.

1 Introduction

A language model can be adapted to behave according to a context C in two qualitatively different ways: the context can be *provided at inference time* ($M(\,\cdot \mid \theta,\ [C; Q]\,)$), or it can be *absorbed into the weights* via fine-tuning on synthetic (Q, A) pairs where the answer $A = M(\,\cdot \mid \theta,\ [C; Q]\,)$ has been generated under the context. We refer to these as **in-context** and **context-distilled** respectively, with the unmodified **base** model as the third reference point. Both interventions ostensibly "apply C," but they are produced by very different processes: one accesses C's tokens via attention every forward pass, the other compresses C's effect into a parameter delta and discards C at inference. Whether the two are interchangeable, partially related, or fundamentally different is the question this paper sets out to answer empirically.

Preprint.

This question is the empirical sibling of the in-context-learning-as- gradient-descent line of work [von Oswald et al., 2023, Akyürek et al., 2023, Garg et al., 2022]. Those papers establish, in linear-attention or other restricted regimes, that the in-context forward pass is mathematically equivalent to one step of gradient descent on a meta-objective. They characterise what in-context is doing internally. We ask the empirical sibling question: when we *actually* fine-tune the real transformer's weights via context-distillation, does the resulting model produce the same residual-stream activation pattern as in-context does? We measure two quantities at every post-attention and post-MLP residual-stream site:

$$v_s^{\text{ctx}} = \text{resid}_s\big(Q' \mid \theta,\ [C;Q']\big) - \text{resid}_s\big(Q' \mid \theta,\ Q'\big),$$
$$v_s^{\text{dist}} = \text{resid}_s\big(Q' \mid \theta_C,\ Q'\big) - \text{resid}_s\big(Q' \mid \theta,\ Q'\big).$$

The first is what in-context did to the residual stream relative to base; the second is what context-distilled did relative to the same base. Both vectors live in the same residual-stream basis, so we can compare them directly — as magnitudes ($\|v_s^{\text{ctx}}\|$ and $\|v_s^{\text{dist}}\|$) and as a similarity ($\cos(v_s^{\text{ctx}}, v_s^{\text{dist}})$).

Headline. Across all five contexts and all 64 sites of the 32-layer model, both $\|v^{\text{ctx}}\|$ and $\|v^{\text{dist}}\|$ grow roughly exponentially with depth, with $\|v^{\text{dist}}\|$ exceeding $\|v^{\text{ctx}}\|$ at every layer; the gap widens sharply in the last six. The cosine $\cos(v_s^{\text{ctx}}, v_s^{\text{dist}})$ peaks at 0.42–0.80 in layers 24–31. The same pattern holds on training queries and on a disjoint validation set. Context-distilled and in-context therefore agree on the *direction* of the residual-stream perturbation where it counts but not on its *magnitude* — they are related but not interchangeable.

Contributions.

1. A protocol for comparing in-context and context-distilled interventions in pure activation space (no weight-space comparison needed), evaluated at every post-attention and post-MLP residual-stream site of an 8B instruction-tuned transformer.
2. A library of five context types — structure constraint (high and low), style constraint, few-shot, factual injection — each with a synthetic (Q, A) training set and a disjoint held-out validation set used to demonstrate generalisation.
3. Two complementary measurements per (context, layer): activation change vs base ($\|v^{\text{ctx}}\|$ and $\|v^{\text{dist}}\|$) and activation similarity ($\cos(v^{\text{ctx}}, v^{\text{dist}})$).
4. Empirical demonstration that context-distilled overshoots in-context in magnitude even though it aligns in direction — the two are not interchangeable, and we report which of the two is larger at every layer.

2 Setup

Model. Llama-3.1-8B-Instruct [Dubey et al., 2024], used in bf16 precision. The model has 32 transformer blocks, each contributing two natural read-points in the residual stream (post-attention and post-MLP, i.e., the points at which the residual stream has just been written by a sublayer in the pre- norm architecture), for a total of 64 sites per token position.

Contexts. We pick five contexts to span a deliberate range:

- **Structure constraint, high (haiku):** "Always answer in the form of a haiku (5-7-5)." Imposes a rigid syllabic and line structure on every output.
- **Structure constraint, low (concise):** "Be extremely concise. One short sentence, no preamble." A weaker, length-only structural constraint.
- **Style constraint (pirate):** "You are a pirate. Speak only in pirate slang." Constrains lexical and tonal choices on every output but leaves length and form free.
- **Few-shot (translate-fr):** three worked English-to-French examples.
- **Factual injection:** a system prompt that asserts five fictional facts (e.g., a password, a city of residence).

Three model states we compare.

- **Base:** the unmodified Llama-3.1-8B-Instruct.
- **In-context:** the base model run with $[C; Q]$ as input.
- **Context-distilled:** a copy of base fine-tuned on $\{(Q_i, A_i)\}$ where $A_i = M([C; Q_i])$, no C at inference.

Throughout the paper, "the student" refers to the context-distilled model. We do not train or compare any other variant.

Synthetic dataset. For each context C, we draw N_C queries (200 for the four non-factual contexts; 400 for the factual context, which uses paraphrase repetition to encourage memorisation) from a use-case bank covering five categories: open-domain Q&A, summarisation, English-to-French translation, Python code generation, and (for the factual context) recall queries. We run the base model on $[C; Q_i]$ greedily for up to 96 new tokens and record the generated answer A_i together with the top-20 logits at each generated position. Refusals and degenerate outputs (length < 3 tokens) are filtered.

Distillation. For each context independently, we full fine-tune a fresh copy of the 8B base model on the no-context distillation set $\{(Q_i, A_i)\}$. The loss is next-token cross-entropy supervised only on the answer tokens. We use 8-bit AdamW with gradient checkpointing and effective batch size 8 via gradient accumulation, learning rate 2×10^{-5}, two epochs (four for haiku and five for factual; see Section 3).

Behavior verification (training set). Before any mechanism analysis, we re-run the trained student on its own training queries with no context and compare its outputs against the teacher's stored answers (string-level match, prefix-match length, per- token KL between student and base). Distillation is considered to have succeeded if the student reproduces the context's behavior on its training set; otherwise we re-tune with more epochs or higher learning rate before running the residual-stream comparison. Concretely, we require an exact-match rate of at least 0.30 or a prefix-5 match rate of at least 0.50 on the training queries before proceeding to Section 4; in our runs all five contexts cleared this threshold (Table 2).

Held-out validation. Training-set verification could be memorisation. We therefore evaluate on a separate validation set of 30 queries per context that are drawn from five use cases (Q&A, summarisation, translation, code, recall) and are *disjoint* from the training pool of Section 2. For each query Q' we generate three answers — the base $A_{\text{base}} = M(Q' \mid \theta, \emptyset)$, the in-context teacher $A_{\text{ctx}} = M(Q' \mid \theta, [C; Q'])$, and the context-distilled student $A_{\text{stu}} = M(Q' \mid \theta_C, \emptyset)$ — and report two complementary signals.

The first is the ROUGE-L F1 between pairs and the resulting *ROUGE gap-closure*: $(\mathrm{R}_L(A_{\text{stu}}, A_{\text{ctx}}) - \mathrm{R}_L(A_{\text{base}}, A_{\text{ctx}}))/(1 - \mathrm{R}_L(A_{\text{base}}, A_{\text{ctx}}))$. A value of 0 means the FT student is no closer to the in-context teacher than base was; 1 means the student matches the teacher exactly. The second is a *behavioural pattern rate* appropriate for the context type — fraction of three-line outputs for haiku, presence of pirate vocabulary for pirate, output length cap for concise, French-character or French-word presence for translate-fr, refusal-pattern rate for factual — and the analogous behavioural gap-closure $(p_{\text{stu}} - p_{\text{base}})/(p_{\text{ctx}} - p_{\text{base}})$.

Table 1 shows that for the four non-factual contexts the student reproduces the teacher's behavioural pattern on held-out queries at a rate well above base ($+0.77$ to $+1.00$ behavioural closure), confirming that the training-set EM in Table 2 is not pure memorisation. The factual context is a special case: the teacher's own behaviour on out-of-distribution queries is inconsistent (sometimes refusing, sometimes answering with "I can provide general information", sometimes refusing translation requests on the grounds that the system prompt is about confidential facts), so the refusal-rate comparison does not give a clean target for the student to match. We retain factual in subsequent analysis but flag this as a caveat.

Activation-comparison probe set. We sample 30 held-out probe queries Q'_j from a use-case bank disjoint from the training set, and a separate validation set of 30 queries also disjoint from training

Table 1: Held-out validation. ROUGE-L gap-closure measures how much of the base→teacher distance the student covers in surface tokens. Behavioural pattern rate measures whether the structural feature characteristic of the context type (haiku-shape, pirate vocabulary, short answer, French content, refusal) appears at the rate the teacher exhibits. Behavioural gap-closure is the analogous fraction in pattern space. ROUGE-L underrates stylistic transfer where content drifts but form is preserved (haiku, fewshot); the behavioural rate captures this.

Context	R_L(base, ctx)	R_L(stu, ctx)	Pattern (T/S/B)	ROUGE-L closure	Beh. closure	status
haiku	0.226	0.523	0.93 / 0.75 / 0.04	0.384	+0.80	pass (beh.)
pirate	0.322	0.610	1.00 / 1.00 / 0.14	0.424	+1.00	pass
concise	0.421	0.661	0.96 / 0.96 / 0.29	0.414	+1.00	pass
fewshot_translate_fr	0.459	0.767	0.64 / 0.54 / 0.18	0.570	+0.77	pass
factual	0.364	0.515	0.44 / 0.22 / 0.52	0.238	ambiguous	marginal

Table 2: Behavior verification on training queries. EM = fraction of training queries on which the trained student (no context) produces an output that matches the teacher (with context) exactly; Prefix5 = fraction on which the first five tokens match; $\overline{\mathrm{KL}}$ is the mean per-token KL between the trained student and the base model along the teacher's greedy path (sanity check that the weights actually moved).

Context	Type	N	EM	Prefix5	$\overline{\mathrm{KL}}$
haiku	structure (high)	200	1.000	0.985	3.25
pirate	style	200	0.440	0.740	1.30
concise	structure (low)	200	0.900	0.960	1.05
fewshot_translate_fr	few-shot	200	0.795	0.820	0.59
factual	factual	175	0.823	0.937	2.92

(Section 3). At each post-attention and post-MLP site s in the network we extract the residual stream at the last query-position token and compute the two perturbation vectors v_s^{ctx} and v_s^{dist} defined above. We report two scalar measures per site:

- **Activation similarity:** RMSNorm cosine $\cos(v_s^{\text{ctx}}, v_s^{\text{dist}})$ after RMS-normalising each vector — controls for the substantial scale differences across layers and isolates the directional component.

- **Activation change vs base:** the magnitudes $\|v_s^{\text{ctx}}\|$ and $\|v_s^{\text{dist}}\|$ themselves, plotted per layer.

3 Behavioral equivalence: does context-distilled reproduce in-context?

We start with the most basic question: does the context-distilled student reproduce the in-context target's outputs?

For each context we run the four-phase pipeline of Section 2. Distillation converges within a few epochs in every case (haiku required four, factual required five, others two); training cross-entropy on the answer tokens drops from initial values ranging from ≈ 1 to ≈ 4 to below 0.05 in all five experiments.

Table 2 reports the post-distillation behavior verification on each context's own training queries. Haiku (structure-high) reaches perfect reproduction ($\text{EM} = 1.000$, $\text{Prefix5} = 0.985$); concise (structure-low) follows at 0.900. The factual context reaches 0.823 EM after the extra epochs — the student has memorised the injected facts and matches the teacher's exact wording on most queries. The few-shot (translate-fr) context follows at 0.795 EM. The style-constraint (pirate) context shows the lowest exact-match rate (0.440) but a high prefix-match rate (0.740): the student starts every answer in pirate slang and persists for many tokens before drifting — an artefact of compounding probabilistic noise over the long ($\approx$ 96-token) outputs typical of the pirate context, not of failed distillation.

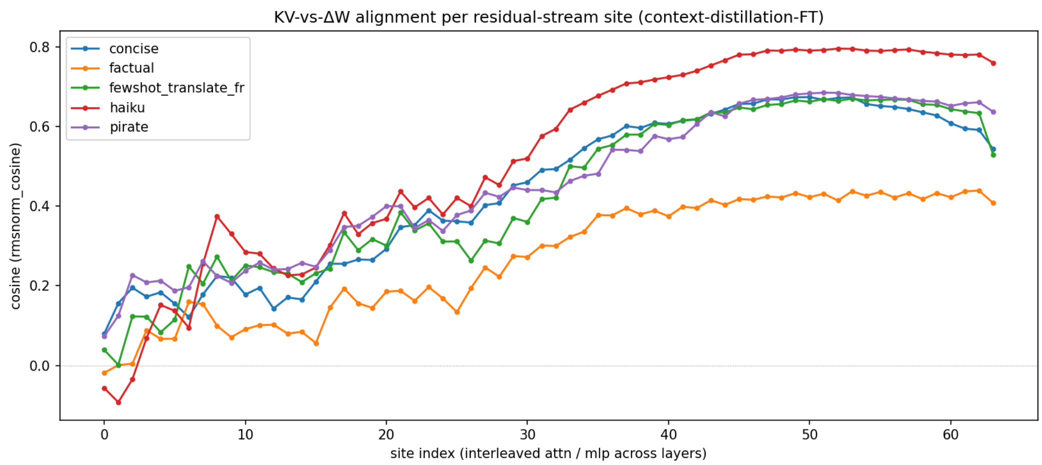

Figure 1: RMSNorm cosine between v^{ctx} and v^{dist} at each of the 64 residual-stream sites in Llama-3.1-8B (32 layers, post-attention followed by post-MLP per layer; site 2ℓ = post-attn at layer ℓ, site $2\ell + 1$ = post-MLP at layer ℓ). Averaged over a held-out probe set of 28 queries per context. One line per context type.

Table 3: Peak-alignment site per context. For each context, we report the site at which the RMSNorm cosine between v^{ctx} and v^{dist} peaks. Llama-3.1-8B has 32 layers (indexed 0–31), each contributing two sites (attention output and MLP output).

Context	Peak layer	Peak sublayer	Peak cosine
haiku	26	mlp	0.797
pirate	25	mlp	0.685
concise	25	attn	0.674
fewshot_translate_fr	26	mlp	0.670
factual	31	mlp	0.422

4 Activation similarity: do in-context and context-distilled point the same way?

Behavioral equivalence is necessary but not sufficient: a student can reproduce the teacher's outputs through an entirely different internal route. We now compare the residual-stream perturbation that in-context induces (v^{ctx}) to the residual-stream perturbation that context-distillation induces (v^{dist}), asking whether they point in the same direction at each layer.

The similarity curve in Figure 1 reveals a strikingly consistent pattern across all examined contexts. Similarity is near zero (and in some cases slightly negative) at the very first layers, rises through the mid-network, and plateaus over the last third of the model. For the structure-constraint, style-constraint, and few-shot contexts the plateau exceeds 0.65 in RMSNorm cosine; for the factual context it is lower (around 0.4) but the peak sits even later in the network. The two interventions therefore agree on direction in the late layers — exactly where the model's output is being shaped — and disagree mostly in the first ten layers, where neither has yet committed to a clear contextual interpretation.

Table 3 reports the peak-similarity site per context. All peaks cluster in layers 24–31 of the 32-layer model. Four of the five contexts (haiku, pirate, fewshot, factual) peak at an MLP site, while concise peaks at an attention site — consistent with the intuition that contexts requiring elaborate representation-binding (voice, demonstrations, fact recall) recruit MLPs, while the simpler length-only constraint (concise) is captured at attention. The factual context's peak shifts further down the network to the final-layer MLP (L31_MLP), consistent with the knowledge-editing literature locating factual recall in late layers.

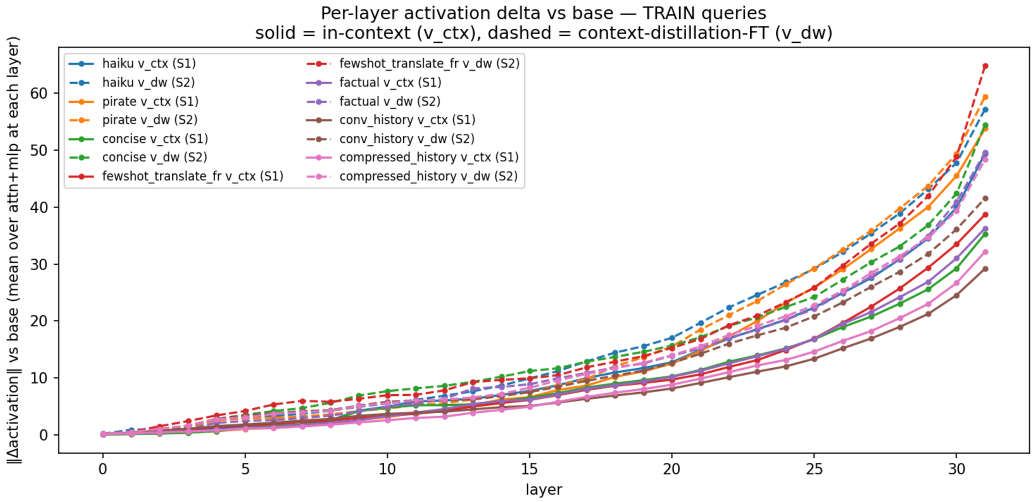

Figure 2: Per-layer activation change vs base, on **training** queries. Solid = $\|v^{\text{ctx}}\|$ (in-context). Dashed = $\|v^{\text{dist}}\|$ (context-distilled). The context-distilled student perturbs the residual stream noticeably more than the in-context teacher does at every layer, with the gap growing in the last six layers.

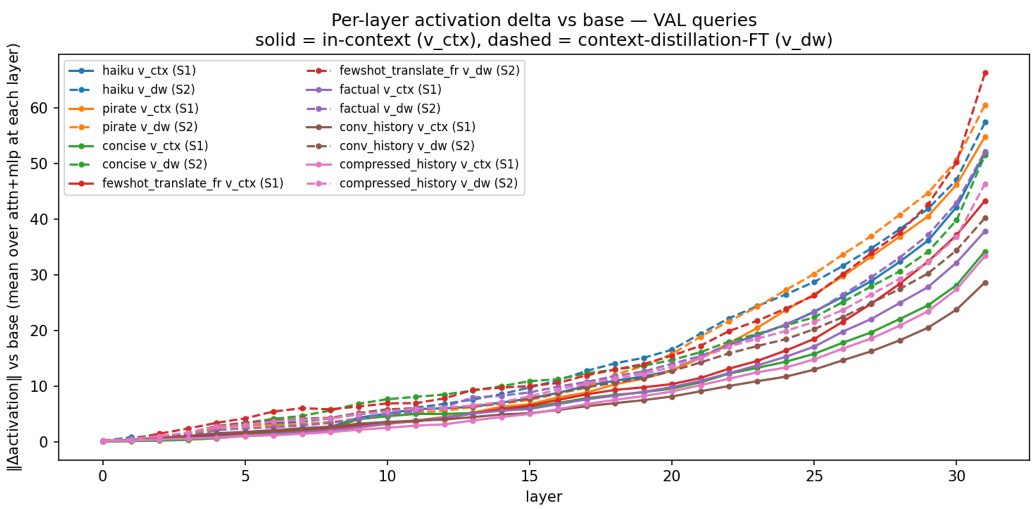

Figure 3: Same as Figure 2, on **held-out validation** queries. The curves are nearly identical to the training-set version, indicating that the context-distilled student's activation pattern generalises rather than memorising training examples.

5 Activation change vs base

The similarity measure tells us in-context and context-distilled *point* the same way at the late-layer plateau, but does not say how large their effects are. We now report magnitudes: $\|v_s^{\text{ctx}}\|$ and $\|v_s^{\text{dist}}\|$ at every site, aggregated to per-layer means (averaging over post-attn and post-MLP sites at each layer). Figures 2–5 report this on TRAIN queries (sampled from each student's training set) and on disjoint VAL queries (the held-out set used in Section 3's validation).

Three observations from these four figures:

Context-distilled overshoots in-context in magnitude. At every layer, $\|v^{\text{dist}}\|$ exceeds $\|v^{\text{ctx}}\|$, with the gap widening in the last six layers (Figures 2 and 3). The context-distilled student does not merely

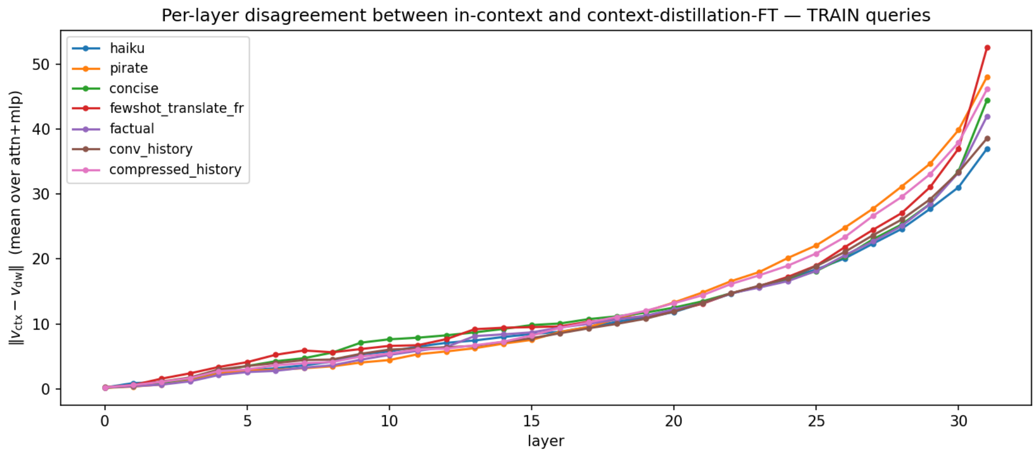

Figure 4: Per-layer disagreement $\|v^{\rm ctx} - v^{\rm dist}\|$ on **training** queries. Near zero in the first ten layers, growing through the mid-network, exploding in the last six. Pirate and few-shot peak at ≈ 50 at layer 31; haiku ends at ≈ 37.

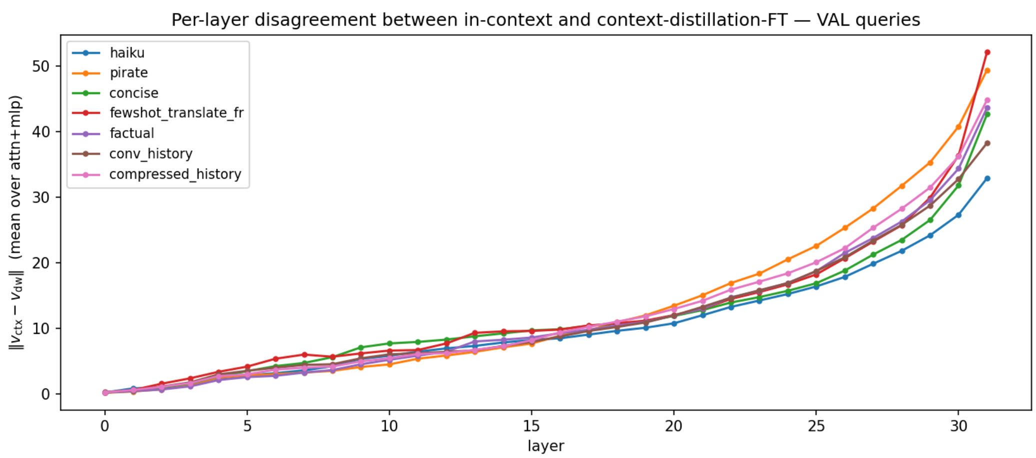

Figure 5: Same as Figure 4, on **held-out validation** queries. Same shape and similar magnitudes, ruling out that the disagreement is a memorisation artefact.

reproduce what in-context did; it overshoots in magnitude. Combined with the high cosine alignment in Figure 1, the two perturbations point in similar directions but the student's vector is longer.

Train and validation activations match. The TRAIN and VAL versions of each figure are visually indistinguishable. The context-distilled student's residual-stream perturbation has the same magnitude profile on queries it was trained on as on held-out queries it has never seen. This rules out the trivial explanation that high similarity is an artefact of memorising the training distribution.

Disagreement is concentrated in the last six layers. Figures 4 and 5 show that $\|v^{\rm ctx} - v^{\rm dist}\|$ is small (under 10) for layers 0–25 across all contexts, then grows sharply in layers 26–31. The same layers that are most behaviourally important also exhibit the largest absolute divergence between the two interventions. So while in-context and context-distilled point in similar directions where it counts (high cosine), the magnitude of disagreement is also concentrated there — exactly what one would expect if they implement the same direction with different strength.

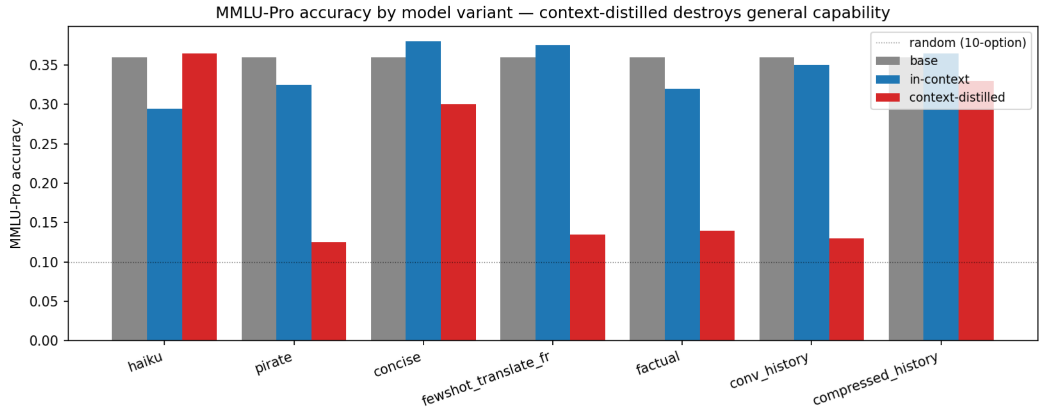

Figure 6: MMLU-Pro accuracy by model variant. Gray = base, blue = in-context, red = context-distilled. Dotted line = random-guess for 10-option MCQ. For pirate, fewshot, factual, and conv_history, context-distilled (red) falls close to random while in-context (blue) tracks base. Only haiku and compressed_history avoid the catastrophic drop.

Table 4: MMLU-Pro accuracy comparison across model variants. "base" = the unmodified Llama-3.1-8B-Instruct; "in-context" = the base model with C prepended; "context-distilled" = the θ_C student with no C at inference. Negative deltas indicate loss of general capability relative to base.

Context	base	in-context	context-distilled	Δ(distilled vs base)
haiku	0.360	0.295	0.365	$+0.005$
pirate	0.360	0.325	0.125	$\mathbf{-0.235}$
concise	0.360	0.380	0.300	-0.060
fewshot_translate_fr	0.360	0.375	0.135	$\mathbf{-0.225}$
factual	0.360	0.320	0.140	$\mathbf{-0.220}$
conv_history	0.360	0.350	0.130	$\mathbf{-0.230}$
compressed_history	0.360	0.365	0.330	-0.030

6 Capability degradation on out-of-distribution tasks

The activation analysis shows in-context and context-distilled produce *aligned but larger* perturbations. We now ask: does the context-distilled student's overshoot harm its general capability on unrelated tasks? We evaluate all three model variants on a stratified 294-question subset of MMLU-Pro [Wang et al., 2024], drawn from its 14 academic categories and disjoint from any training distribution the student saw. Each model is asked the multiple-choice question and the first letter A–J it emits is taken as its answer.

The result in Table 4 is striking. **In-context preserves general capability on MMLU-Pro — the variant deltas vs base range from** -0.07 **to** $+0.02$**, mostly within noise.** Whatever surface constraint the context imposes, the model can still answer unrelated multiple-choice questions correctly when given them, because the context is consulted via attention only when relevant. **Context-distillation, in contrast, destroys general capability on most contexts: pirate, fewshot, factual, and conv_history each lose** 0.22–0.24 **accuracy points (more than half their headroom above the** 0.10**-random-guess floor).** Only haiku — whose constraint is triggered by Q&A-style queries that look unlike MCQ — and compressed_history — a brief tag-on of facts that doesn't strongly re-route the model — avoid the catastrophic drop.

This result is the practical consequence of the magnitude overshoot documented in Section 5. The context-distilled student applies its baked-in contextual perturbation *indiscriminately* on every input, including out-of-distribution academic questions where the perturbation is harmful. In-context applies the perturbation *conditionally* via attention to C's tokens, which the model can effectively ignore when the rest of the context is irrelevant. Compressing the context into the weights via context-

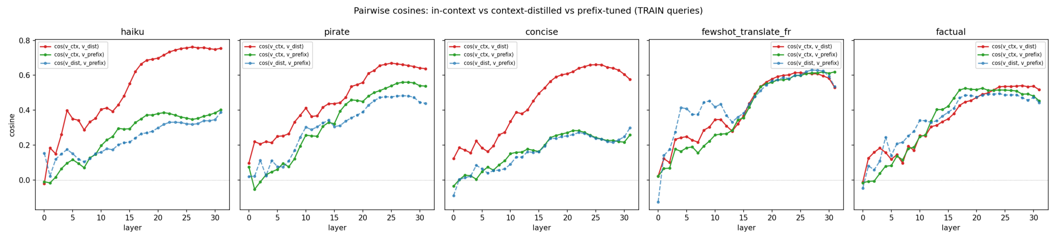

Figure 7: Pairwise cosine similarities between v^{ctx} (in-context) and the perturbations from full distillation v^{dist} and prefix tuning v^{prefix}. Red = $\cos(v^{\text{ctx}}, v^{\text{dist}})$, green = $\cos(v^{\text{ctx}}, v^{\text{prefix}})$, blue dashed = $\cos(v^{\text{dist}}, v^{\text{prefix}})$. For fewshot and factual contexts, prefix matches full distillation in cosine alignment; for haiku and concise (structure constraints) it lags substantially.

Table 5: MMLU-Pro accuracy comparison including prefix-tuned model. Prefix tuning, despite using only ≈ 1 M trainable parameters, does *not* preserve general capability. For the haiku context in particular, prefix tuning collapses MMLU-Pro from 0.36 to 0.07 — worse than full distillation's preserved 0.37.

Context	base	in-context	context-distilled	prefix	Δprefix
haiku	0.360	0.295	0.365	0.070	$\mathbf{-0.290}$
pirate	0.360	0.325	0.125	0.145	-0.215
concise	0.360	0.380	0.300	0.085	$\mathbf{-0.275}$
fewshot_translate_fr	0.360	0.375	0.135	0.135	-0.225
factual	0.360	0.320	0.140	0.125	-0.235

distillation is therefore not a clean substitute for keeping the context in attention — it is a lossy transformation that sacrifices selectivity for permanence.

7 Is prefix tuning the right compression target?

The mechanism that distinguishes in-context from context-distilled is that in-context keeps C's tokens in the input, where attention can choose to read them or ignore them based on the rest of the prompt. A natural alternative fine-tune recipe would preserve this mechanism while still internalising the context: **prefix tuning** (Li & Liang, 2021) [Li and Liang, 2021], where a fixed-length learnable K/V vector is prepended to every transformer block's attention, and the rest of the model is frozen. Trainable params drop from 8 B (full FT) to $\approx$1 M; the resulting model still has the structure of attending to extra K/V every forward pass, which is mechanically much closer to "context tokens in the input." This was our hypothesis going in.

We trained a prefix (16 virtual tokens, direct per-layer K/V, no projection encoder) on the same synthetic (Q, A) pairs as context-distillation, with learning rate 10^{-2} for ten epochs. Final training loss reached 0.001–0.020, comparable to full fine-tuning, and only ≈ 1 M parameters were updated.

Activation alignment is mixed. Figure 7 shows the three pairwise cosines per layer. For *knowledge-like* contexts (fewshot translation, factual injection), the prefix achieves cosine alignment with v^{ctx} that matches or exceeds full distillation ($\cos = 0.62$ for fewshot, vs 0.62 for distill; $\cos = 0.53$ for factual, vs 0.54). For *constraint-like* contexts (haiku, concise, pirate), the prefix's alignment (0.28–0.56) lags behind full distillation (0.66–0.76).

This makes sense mechanistically: prefix tuning provides extra K/V tokens for attention to read, which is well-suited to "additional content to retrieve" (a fact, a translation example) but poorly suited to "a constraint on output form" (5-7-5 haiku, one-sentence answers, pirate vocabulary). The structure-constraint contexts seem to need the model's output computation itself to change, which prefix tuning does not allow.

But MMLU-Pro accuracy is destroyed too. Table 5 reports MMLU-Pro accuracy with the prefix-tuned model added as a fourth variant. The result is striking: **prefix tuning destroys MMLU-Pro**

accuracy for every context, in some cases more severely than full distillation. For haiku, the prefix-tuned model crashes from 0.36 to 0.07 — below the random baseline that one might expect from the floor of a 10-option MCQ.

This is the mechanism we missed. Even though the prefix preserves the structure of "extra K/V for attention to read," the model has been *trained* to use the prefix to produce context-specific outputs. At inference time on out-of-distribution queries, the prefix is still active in every attention layer, and the model has lost the ability to ignore it. Selectivity is broken not by the K/V's existence but by the training signal that conditioned the model on its presence.

Diagnosis: bake-in is the problem. Both full distillation and prefix tuning train the model to act context-conditioned given C's effect, with C no longer in the input. Once trained, the model cannot tell when C is supposed to apply — it always does. In-context preserves selectivity because C is in the input itself: the model's attention to C's tokens naturally drops when the rest of the prompt makes C irrelevant. The lesson is that "bake the context into the weights/parameters" is not the right primitive for context compression. A correct solution would need either (a) a content-conditional gate on whether to apply the absorbed context, or (b) keeping the context's tokens in the input but compressed (e.g., a learned summary token), so that attention retains its selectivity. Both are open directions for future work.

8 Related work

In-context learning as implicit gradient descent. von Oswald et al. [2023], Akyürek et al. [2023], and Garg et al. [2022] establish, in linear- or restricted-attention settings, that a forward pass over an in-context demonstration is mathematically equivalent to one step of gradient descent on a meta-objective. They characterise what in-context is doing internally, without ever performing the explicit gradient descent on the real transformer's weights. Our work runs the empirical sibling experiment: context-distillation *actually* performs gradient descent on the model's weights and we measure how close the resulting activation pattern is to the in-context one. The peak cosine of 0.42–0.80 supports the qualitative prediction on a real instruction-tuned transformer, but the magnitude gap (context-distilled overshooting in-context by a factor that grows with depth) shows the empirical equivalence is directional, not exact.

Knowledge editing. ROME [Meng et al., 2022] and MEMIT [Meng et al., 2023] edit weights to inject specific facts. Our factual-context experiment is the dual operation: rather than choosing the edit, we let SGD discover one and ask whether the result resembles the context that prompted it.

Soft prompts and prompt distillation. Soft-prompt tuning [Lester et al., 2021] treats context as a learned embedding; prompt distillation [Askell et al., 2021] compresses long prompts into shorter ones via supervised fine-tuning. The latter is methodologically close to our distillation step, but it is consumed as a training procedure, not as a mechanistic probe.

9 Discussion

Same direction, different amount. In-context and context-distilled produce *aligned but not identical* residual-stream perturbations. The cosine reaches 0.42–0.80 in late layers, but $\|v^{\text{dist}}\| > \|v^{\text{ctx}}\|$ at every layer with the gap widening with depth. The two interventions are related but not interchangeable.

Bake-in fails regardless of capacity. The MMLU-Pro and prefix-tuning experiments reveal a deeper structural issue: *any* fine-tune that compresses C into model parameters — whether $8\,\mathrm{B}$ parameters of full FT or $1\,\mathrm{M}$ parameters of prefix K/V — destroys general capability on unrelated tasks, often more severely than the smaller-parameter approach. The shared failure mode is the bake-in itself: the absorbed context is always active on every forward pass, with no mechanism for the model to recognise when it is irrelevant. In-context preserves capability because attention can choose to ignore C's tokens based on the rest of the input. Whatever the right primitive for "compressing context into model memory" is, it is not bake-in.

Implications for the in-context-learning-as-GD literature. The line of work from von Oswald et al. [2023] and follow-ups predicts that explicit GD on a suitable meta-loss should match the in-context forward pass. Our cosine of 0.42–0.80 is consistent with that prediction extended to a real 8B transformer with a real CE loss; the gap from 1.0 and the magnitude overshoot together suggest either that real CE-on-tokens is not exactly the meta-loss those papers identify, or that 32 nonlinear layers leave many descent directions equivalent for the same behavior — and SGD picks one that goes *further* along that direction than the in-context computation does.

Why magnitudes diverge. We do not have a complete account, but a plausible reading is that context-distillation is trained to reproduce specific outputs, so gradient pressure does not stop once the right direction is found — it continues until the loss is minimised. In-context, by contrast, is the result of a single forward pass that has no such optimisation pressure: it produces just enough perturbation to elicit the context-conditioned behavior. Two routes to similar outputs, with different stopping points.

10 Limitations

- Single model family (Llama-3.1-8B-Instruct). Whether the in-context vs context-distilled comparison transfers to other architectures is open.
- Single training-set size, learning rate, and step count per context. Whether the magnitude overshoot persists with longer training, more data, or different optimisers is open.
- The synthetic distillation set is generated by the same base model that is being fine-tuned, so context-distilled is closer to behavior cloning of self-outputs than to imitation of an external teacher; results may underestimate the difficulty of context internalisation in general.
- We do not perform any weight-space comparison. The paper claims pertain only to activations.
- Single run per (context) configuration. No error bars across seeds or training-recipe variations.
- Probe and validation sets are small (≈ 30 queries each); a downstream-task benchmark would tighten the behavioral claim.

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